

The Irreversible Blind Spot: Why Symmetric Self-Supervised Objectives Cannot See the Arrow of Time

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Abstract

The companion paper *Beyond the Conductor Blind Spot* showed that the standard self-supervised recipe — align positive pairs, keep the embedding whitened — recovers, at its population optimum, the slowest eigenfunctions of the pair-generating transition operator T , and that a *linear* reading of that representation is complete exactly on the Gaussian stratum. That analysis, like the conductor blind spot it generalises, lives on one axis: the **cumulant** axis, where an order-two instrument cannot carry an order-three fact. This note exhibits a second, independent axis of the same phenomenon — the **reversibility** axis — and the architecture that resolves it.

We prove that the single-encoder alignment objective is a functional of the **symmetric part** $S = \frac{1}{2}(T+T^*)$ of the transition operator *alone*. The antisymmetric part $A = \frac{1}{2}(T-T^*)$ — which is nonzero exactly when detailed balance fails, and which *is* the probability current, the arrow of time — lies in the **kernel** of the objective. The consequence is exact and falsifiable: a world and its time-reverse produce a bit-for-bit identical loss landscape, so no representation trained this way can encode the direction of time, even in principle. This is a blind spot of the same family as the conductor blind spot — a symmetric (second-order) instrument blind to an antisymmetric fact — now on the cycle $\mathbb{Z}/n$ with a current rather than a cumulant. The resolution is structural, not a matter of more data or capacity: the irreversible content is recoverable only by an objective sensitive to $\langle f, Tg \rangle$ with $f \neq g$ — a **predictive (two-encoder)** objective, whose optimum is the singular value decomposition of T . For non-normal dynamics this sharpens into a geometric statement: the **left and right singular charts differ** — the coordinate z is encoded in is not the coordinate z' is read in — and a single nonnegative **irreversibility gap** $\Delta = \sum_i \sigma_i(T) - \sum_i \lambda_i(S) \geq 0$ measures the predictable structure the symmetric objective forfeits, with two faces (a phase deficit for a normal current, a chart split for a non-normal one) and vanishing exactly at detailed balance. The predictor network of BYOL/SimSiam/I-JEPA is, on this reading, not merely an anti-collapse device but the minimal mechanism that lifts the objective out of the irreversible blind spot. Finally the result is **topological**: a stationary current is a divergence-free flow, so the irreversible structure a symmetric objective discards has, on a graph, exactly $\beta_1 = E - V + 1$ independent modes (the *cycle rank*), and in the continuum exactly $\dim H^1(M) = b_1(M)$ — the first Betti number of the state manifold, the number of *harmonic* currents that circulate its holes. The discrete count refines to the continuum one when the mesh is filled (a 3×3 torus's $\beta_1 = 10$ collapses to $b_1(T^2) = 2$); for a learned representation $b_1(M)$ counts the periodic latent factors that carry a real arrow of time. Every claim is settled by deterministic, exact-arithmetic certificates — E9 on the drift ring (the arrow as imaginary spectrum), E10 on the non-normal conveyor (the arrow as a 60° chart split), and E12 across graphs of growing cycle rank and their Hodge Betti numbers (the arrow's dimension as β_1 , refining to b_1) — all showing the blind spot switch off precisely at detailed balance. Two further certificates address practice: E13 finds the arrow's *strength* cheaply and robustly estimable from finite samples while its *dimension* b_1 is an offline topological audit (exact persistence times out; single-scale estimates are wrong), E14 finds the reversibility axis and the parent's cumulant axis **orthogonal** — independently dialable, jointly auditable; E15 makes the topological audit efficient — b_1 as the

nullity of the sparse Hodge 1-Laplacian, exact and scaling to thousands of simplices sub-second; and E16 closes the loop, recovering b_1 from raw point clouds by persistent homology and feeding E15's engine, so the audit runs end-to-end.

Keywords: self-supervised learning, transition operator, detailed balance, time reversal, probability current, predictor networks, slow feature analysis, arrow of time, identifiability, singular value decomposition, Hodge/Helmholtz decomposition, graph homology, Betti number.

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1. Two axes of one blindness

The conductor blind spot is a statement on the **cumulant axis**: on a categorical head with cyclic structure, the order-two Fisher form is constitutionally unable to carry the order-three Amari–Chentsov cubic. The parent paper carried that boundary into continuous representation learning: the whitened-alignment optimum recovers the slow-eigenfunction chart of a world, and a *linear* probe of it is complete only on the Gaussian (second-order-sufficient) stratum. Both statements are about the *order* at which a second-order class stops sufficing.

There is a second axis, orthogonal to the first, and the parent paper named it only to defer it: its §8 lists **reversibility** (detailed balance) as the structural hypothesis that “buys the self-adjoint spectral theorem,” and leaves “genuinely non-reversible additive noise” to follow-up. This note is that follow-up, and the deferral turns out to hide a blind spot exactly as sharp as the first.

The object is the same cycle. Take the conductor blind spot's $\mathbb{Z}/n$ and give it a **net drift**: a walk that steps forward with probability q and backward with probability $b \neq q$. Detailed balance is broken; the walk carries a **probability current**, a preferred direction — an arrow of time. We ask the question the parent paper asks of every world: *what does the aligned, whitened encoder recover?* The answer is that it recovers everything the parent paper says it does — and is **exactly blind to the arrow**. The reason is a one-line fact about symmetric forms, and the cure is a specific, familiar piece of architecture.

1.1 The smallest irreversible world

On $\mathbb{Z}/n$ let the transition operator be

$$(T\psi)(k) = q\psi(k+1) + b\psi(k-1) + r\psi(k), \quad r = 1 - q - b,$$

with the uniform law stationary (it is doubly stochastic). In L^2 of the uniform law the adjoint is the matrix transpose, so T splits as

$$T = S + A, \quad S = \frac{1}{2}(T + T^*) \text{ (forward = backward = } \frac{q+b}{2}\text{)}, \quad A = \frac{1}{2}(T - T^*) \text{ (the current).}$$

S is a plain lazy random walk on the ring — no arrow. $A\psi(k) = \frac{q-b}{2}(\psi(k+1) - \psi(k-1))$ is a centred difference, the discrete derivative, **nonzero exactly when $q \neq b$** . The slow eigenvalue is complex,

$$\lambda_1 = \underbrace{r + (q+b)\cos\frac{2\pi}{n}}_{\alpha = \text{Re } \lambda_1} + i \underbrace{(q-b)\sin\frac{2\pi}{n}}_{\beta = \text{Im } \lambda_1},$$

and the entire arrow of time is the single number β , whose sign is the drift direction. Hold this picture: **the real part α is what the standard objective will see; the imaginary part β is what it will throw away.**

2. The Irreversible Blind Spot

We use the parent paper’s reduction verbatim. With $\{h_i\}$ whitened and T the transition operator, stationarity gives $\mathbb{E}[h_i(z')h_i(z)] = \langle h_i, Th_i \rangle$, so (parent, Appendix A.1)

$$\mathbb{E}\|h(z') - h(z)\|^2 = \sum_i 2(1 - \langle h_i, Th_i \rangle). \quad (2.1)$$

Lemma 1 (skew forms vanish). *For the $L^2(p)$ adjoint, $A = \frac{1}{2}(T - T^*)$ is skew-adjoint ($A^* = -A$), so for every real $h \in L^2(p)$, $\langle h, Ah \rangle = 0$. Proof. $\langle h, Ah \rangle = \langle A^*h, h \rangle = -\langle Ah, h \rangle = -\langle h, Ah \rangle$, and a real number equal to its own negative is 0. $\square$*

Theorem 1 (Irreversible Blind Spot). *The single-encoder whitened-alignment objective (2.1) is a functional of the symmetric part S alone:*

$$\mathbb{E}\|h(z') - h(z)\|^2 = \sum_i 2(1 - \langle h_i, Sh_i \rangle). \quad (2.2)$$

Consequently:

1. (**Time-reversal degeneracy.**) *The map $T \mapsto T^*$ (reverse the pair, flip the arrow) leaves S fixed, hence leaves the objective unchanged as a functional of h . A world and its time-reverse have a **bit-for-bit identical loss landscape**: every encoder receives the same loss, so every minimiser — and the entire recovered representation — coincides.*
2. (**The arrow is in the kernel.**) *$\partial/\partial A$ of the objective is identically zero: no perturbation of the antisymmetric part changes any value of the loss. Since $A \neq 0$ exactly when detailed balance fails, and A is the generator of the probability current, the irreversible content — the arrow of time — is unrecoverable by any single-encoder representation trained this way.*

Proof. (2.2) is (2.1) with $\langle h_i, Th_i \rangle = \langle h_i, Sh_i \rangle$ by Lemma 1. Both consequences are immediate: S is the symmetric part of both T and T^* , and the right-hand side of (2.2) does not contain A . $\square$

Corollary 1 (what is recovered instead). *Under a slow-feature gap on S , the minimiser is $h = U\Psi_1$ with U orthogonal and Ψ_1 the top eigenspace of S — the slow chart of the **reversibilised** dynamics $\frac{1}{2}(T + T^*)$, the closest reversible world to T . The encoder recovers the world with its arrow erased. (This is exactly the parent paper’s Theorem 1, with S in place of T ; everything the parent proves about the chart’s curvature and the Gaussian boundary applies to S unchanged. The cumulant axis and the reversibility axis are independent: S can still be non-Gaussian, so the chart can still be curved — the two blind spots compose.)*

This is the conductor blind spot’s structure on a new axis. There, an order-two instrument (the Fisher form) was blind to an order-three fact (the cubic). Here, a symmetric instrument (a quadratic

form in a single encoder) is blind to an antisymmetric fact (the current). Both are second-order classes failing to carry a piece of the world’s geometry that is, by a parity/symmetry argument, simply not in their range.

3. The resolution is a predictor

The blind spot is a property of the form $\langle h, Th \rangle$, which symmetrises T before the encoder ever sees it. To recover A the objective must probe T off the diagonal — it must evaluate $\langle f, Tg \rangle$ for $f \neq g$. That is precisely what a **predictive** objective does.

Fix a whitened embedding h and introduce a learnable linear **predictor** P acting in embedding space, trained to predict the partner’s embedding:

$$P^* = \arg \min_P \mathbb{E} \|Ph(z) - h(z')\|^2 = \mathbb{E}[h(z') h(z)^\top] \quad (\text{whitened } h),$$

whose entries are $P_{ab}^* = \mathbb{E}[h_a(z') h_b(z)] = \langle h_b, Th_a \rangle$. So P^* is the **matrix of the operator T itself** in the embedding basis — not its quadratic form.

Theorem 2 (the predictor recovers the arrow). *Restricted to the slow subspace with orthonormal basis $\{\psi_a\}$, the predictive optimum equals the operator block $B_{ab} = \langle \psi_a, T\psi_b \rangle$. Its symmetric part $\frac{1}{2}(B + B^\top)$ is exactly what the single-encoder objective (2.2) can see; its antisymmetric part*

$$\frac{1}{2}(B - B^\top)_{ab} = \langle \psi_a, A\psi_b \rangle$$

is a faithful readout of the current. Under time reversal $B \mapsto B^\top$. In particular a strictly asymmetric optimal predictor, $P^ \neq (P^*)^\top$, exists if and only if detailed balance fails.*

Proof. P^* has entries $\langle \psi_b, T\psi_a \rangle$ by the normal-equations computation above; restricting to the slow subspace gives B . Splitting $T = S + A$ and using $S^* = S$, $A^* = -A$ gives $\frac{1}{2}(B + B^\top)_{ab} = \langle \psi_a, S\psi_b \rangle$ and $\frac{1}{2}(B - B^\top)_{ab} = \langle \psi_a, A\psi_b \rangle$. Time reversal sends $A \mapsto -A$, hence $B \mapsto B^\top$. The asymmetry is nonzero iff $A \neq 0$ iff detailed balance fails. $\square$

Corollary 2 (the architecture the theorem explains). *A symmetric Siamese objective — shared encoder, covariance/whitening anti-collapse, no predictor — is provably arrow-blind (Theorem 1). The predictor head of BYOL, the stop-gradient asymmetry of SimSiam, and the predictor of I-JEPA are the minimal devices that make the objective sensitive to $\langle f, Tg \rangle$ with $f \neq g$, lifting it out of the irreversible blind spot. On temporal and causal data — where the transition is genuinely irreversible — this predicts that predictive architectures capture structure that symmetric contrastive ones cannot, and locates the reason in the antisymmetric part of one operator. We state this as the structural reading the theorem licenses; a controlled head-to-head on irreversible worlds is the natural empirical test.*

3.1 The ring, worked

On the drift ring of §1.1, the slow subspace is spanned by the L^2 -orthonormal pair $\psi_1 = \sqrt{2} \cos \frac{2\pi k}{n}$, $\psi_2 = \sqrt{2} \sin \frac{2\pi k}{n}$, and a one-line computation gives the predictor block in closed form,

$$B = \begin{pmatrix} \alpha & \beta \\ -\beta & \alpha \end{pmatrix}, \quad \alpha = r + (q + b) \cos \frac{2\pi}{n}, \quad \beta = (q - b) \sin \frac{2\pi}{n}.$$

The symmetric part is αI — an isotropic scaling, all the single encoder can read, and identical under $q \leftrightarrow b$. The antisymmetric part is βJ (with $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$) — a rotation generator whose sign is the drift. The predictor turns the static recovered ring into a **rotational flow**, and the direction of that rotation *is* the arrow of time; reversing the drift transposes B and spins the flow the other way. At detailed balance $q = b$ we have $\beta = 0$, $B = \alpha I$ is symmetric, predictor and single encoder agree, and there is no arrow to miss.

3.2 The general (non-normal) form: two charts, and the irreversibility gap

The ring is *normal* ($TT^* = T^*T$), so its arrow is carried by the imaginary spectrum and the predictor block is a clean rotation inside a **single** recovered chart. The general statement drops normality. Let $T = \sum_k \sigma_k u_k \langle v_k, \cdot \rangle_p$ be the singular value decomposition of T on the mean-zero subspace of $L^2(p)$: $Tv_k = \sigma_k u_k$, with $\{v_k\}$ the **right** (input) chart, $\{u_k\}$ the **left** (output) chart, both $L^2(p)$ -orthonormal.

Theorem 3 (general recovery is the SVD; two charts). *The two-encoder objective $\min_{f,g} \mathbb{E} \|f(z') - g(z)\|^2$ over whitened f, g has optimum value $2(d - \sum_{i \leq d} \sigma_i)$, attained at $g_i = u_i$, $f_i = v_i$. The input and output charts span the same subspace if and only if T is normal; for a non-normal T they differ, and the principal angle between $\text{span}\{v_i\}$ and $\text{span}\{u_i\}$ is a coordinate-free measure of irreversibility. Time reversal $T \mapsto T^*$ exchanges the two charts.*

Proof. With f, g whitened, $\mathbb{E} \|f(z') - g(z)\|^2 = \sum_i 2(1 - \langle g_i, T f_i \rangle)$, and maximising the bilinear form $\sum_i \langle g_i, T f_i \rangle$ over orthonormal $\{f_i\}, \{g_i\}$ gives $\sum_{i \leq d} \sigma_i$ at the singular functions (the Eckart–Young / Ky Fan principle for singular values). The left and right singular subspaces coincide iff T is normal; and $T^* = \sum_k \sigma_k v_k \langle u_k, \cdot \rangle_p$ swaps $u \leftrightarrow v$. $\square$

So irreversibility is not one phenomenon but a **chart split**: the coordinate in which z is most predictable is not the coordinate in which z' is read. A single shared encoder, forced to use $f = g$, cannot represent the split — it collapses to the symmetric part (Theorem 1). The two failures, the cumulant axis (parent) and the reversibility axis (here), are independent, and the reversibility axis itself now has two faces: a phase (normal) and a chart split (non-normal).

Both faces are measured by one nonnegative scalar. The single encoder captures $\sum_{i \leq d} \lambda_i(S)$ (Theorem 1 with Ky Fan on S); the two encoder captures $\sum_{i \leq d} \sigma_i(T)$; and by the Fan–Hoffman inequality $\lambda_i(S) \leq \sigma_i(T)$, so

The irreversibility gap. $\Delta := \sum_{i \leq d} \sigma_i(T) - \sum_{i \leq d} \lambda_i(S) \geq 0$ is the predictable correlation the single encoder leaves on the table. $\Delta = 0$ if and only if T is self-adjoint on the top- d subspace (detailed balance there). $\Delta > 0$ arises two ways: a **normal current** ($\sigma_i = |\lambda_i| > \text{Re } \lambda_i = \lambda_i(S)$, the §2 imaginary-spectrum mechanism) or **non-normality** (the chart split, this section). One gap, two mechanisms.

On the drift ring Δ is the phase deficit $\sum (|\lambda_i| - \text{Re } \lambda_i) > 0$; on a conveyor with position-dependent rates it is the chart-split deficit. Both, and the exact collapse $\Delta = 0$ at detailed balance, are certified in E10.

3.3 The topology of the blind spot: $\dim(\text{arrow}) = \text{cycle rank}$

The drift ring carries a *single* current (one β). A general world carries a whole space of them, and its dimension is a topological invariant of the transition graph. This is the deepest form of the result, and it is exact.

Let T act on a connected graph $G = (V, E)$ with stationary law π . The **stationary current** $J_{ij} = \pi_i T_{ij} - \pi_j T_{ji}$ is antisymmetric and, crucially, **divergence-free**: by stationarity $\sum_j J_{ij} = \pi_i \sum_j T_{ij} - \sum_j \pi_j T_{ji} = \pi_i - (\pi^\top T)_i = 0$ (Kirchhoff's current law). The divergence-free edge flows on G are exactly the kernel of the boundary operator $\partial_1 : \mathbb{R}^E \rightarrow \mathbb{R}^V$ — the **cycle space** — of dimension the first Betti number

$$\beta_1(G) = \dim \ker \partial_1 = E - V + 1.$$

Theorem 4 (topological dimension of the irreversible blind spot). *The single-encoder objective is invariant under any change of the current J (Theorem 1: it is a function of S alone), whereas the predictor recovers J in full (Theorem 3). Hence the dimension of the irreversible content invisible to a symmetric objective — the space of currents sharing one reversible backbone S and one stationary law π — equals the cycle rank*

$$\dim(\text{irreversible blind spot}) = \beta_1(G) = E - V + 1.$$

Detailed balance is the origin $J = 0$; the drift ring ($\beta_1 = 1$) is the minimal nonzero case.

Proof. J is antisymmetric by construction and divergence-free as shown, so $J \in \ker \partial_1$, of dimension $E - \text{rank } \partial_1 = E - (V - 1) = \beta_1$ for connected G . Adding a **circulation** — a skew, divergence-free edge flow — to T changes only $A = \frac{1}{2}(T - T^*)$, leaving S and π fixed, so the entire β_1 -parameter family of worlds shares one single-encoder optimum (Theorem 1). The predictor recovers the operator T , hence A , hence J (Theorem 3), so all β_1 directions are recoverable. $\square$

The arrow of time is therefore not a scalar but a **homology class count**: the irreversible structure a symmetric objective discards has exactly as many independent modes as the transition graph has independent cycles. A tree ($\beta_1 = 0$) carries no recoverable arrow; a single loop carries one (E9); a richly connected world — a 3×3 torus *as a graph* has $\beta_1 = 10$ — hides a multidimensional current behind one reversible backbone. Connectivity is thus a design knob for irreversible expressivity: capturing many independent modes of temporal/causal structure requires a transition graph of high cycle rank, and a predictor to read them.

3.4 The continuum: the harmonic arrow ($\dim = \dim H^1$)

Theorem 4 is combinatorial. Its continuum shadow is cleaner and is a genuine topological invariant of the *state space*, not of a chosen mesh. Let the state space be a closed oriented Riemannian manifold M (or $\mathbb{R}^n$ with a confining potential and decay at infinity), with a reversible diffusion backbone of stationary density p and an added stationary current. The probability current $J = bp - D\nabla p$ is a vector field, and stationarity is the continuity equation $\nabla \cdot J = 0$; via the metric, J is a **co-closed** 1-form, $\delta J = 0$. The Hodge decomposition $\Omega^1 = \text{im } d \oplus \text{im } \delta \oplus \mathcal{H}^1$ then forces

$$J = \underbrace{\delta\beta}_{\text{co-exact eddy}} + \underbrace{h}_{\text{harmonic}}, \quad h \in \mathcal{H}^1(M) \cong H_{\text{dR}}^1(M),$$

with no exact part (on a closed manifold $\delta J = 0$ kills it).

Theorem 5 (continuum: the harmonic arrow). *Under the hypotheses above: (i) the single-encoder objective is blind to the entire current J — Theorem 1 is metric/operator-level and uses no graph structure, only the $L^2(p)$ self-adjoint/skew splitting of the generator; (ii) the co-exact part $\delta\beta$ is locally a curl (a contractible eddy, removable by a local change of dynamics), while the harmonic*

part h is the obstruction no local change removes — the current forced to circulate the holes of M ; (iii) the harmonic currents form $\mathcal{H}^1(M) \cong H_{\text{dR}}^1(M)$, of dimension the first Betti number $b_1(M)$. This $b_1(M)$ is the topologically protected, mesh-independent dimension of the arrow, and Theorem 4 is its 1-complex degeneration: a graph has no 2-cells, so every divergence-free flow is harmonic and the blind spot has dimension $\beta_1(G) = E - V + 1$; filling the 2-cells sends the contractible cycles into the $\delta\beta$ eddies and leaves exactly $b_1(M)$ harmonic modes.

Proof sketch. (i) The identity $\langle h, Th \rangle = \langle h, Sh \rangle$ of Theorem 1 is the statement that the skew part of the generator contributes zero to every real quadratic form; it holds verbatim on M . (ii)–(iii) $\nabla \cdot J = 0 \iff \delta J = 0$; a co-closed form has $J = \delta\beta + h$ (the exact part vanishes since $\delta d\alpha = 0 \Rightarrow d\alpha = 0$ on closed M); the harmonic projection is the unique harmonic representative, and the Hodge theorem gives $\mathcal{H}^1(M) \cong H_{\text{dR}}^1(M)$, $\dim = b_1(M)$. The graph case: a 1-complex has trivial co-exact part, so $\ker \delta$ (divergence-free flows) $= \mathcal{H}^1$, dimension β_1 ; the boundary d_2 of added 2-cells enlarges $\text{im } \delta$ and shrinks $\mathcal{H}^1$ to b_1 of the complex. $\square$

The refinement is a *correction*, not a technicality: a discrete model’s cycle rank $E - V + 1$ **overcounts** the arrow, because most of its cycles are discretisation artifacts (the little faces of the mesh) that a finer model fills in. E12 makes this exact — the 3×3 torus graph’s $\beta_1 = 10$ collapses to $b_1(T^2) = 2$ once its square faces are filled, and the tetrahedral K_4 ($\beta_1 = 3$) collapses to $b_1(S^2) = 0$ (a sphere has no holes; every current on it is a removable eddy). The invariant that survives *every* discretisation — the true count of independent irreversible modes — is $b_1(M)$. For a learned representation, M is the low-dimensional data manifold the encoder discovers, and $b_1(M)$ counts the **periodic/cyclic latent factors** — rotation, phase, time-of-day, gait, conversational turn-taking — that carry a genuine arrow of time and that a symmetric objective is, by Theorem 1, constitutionally unable to see.

Remark (estimating the harmonic arrow without meshing). Theorem 5 appears to demand an expensive object — a Hodge decomposition of a high-dimensional space — but in practice none is built, for three reasons. (1) *The chart is already low-dimensional.* The decomposition is read in the learned embedding $\mathbb{R}^d$, not the ambient data space; the encoder has done the dimensionality reduction, and b_1 of the slow data manifold is small. One never meshes the ambient space (which would incur the curse of dimensionality and defeat any persistent-homology approach). (2) *The predictor already estimates the current.* The two-encoder optimum $P^\star = \mathbb{E}[f(z')f(z)^\top]$ (whitened) is the operator in the embedding basis; its antisymmetric part $\frac{1}{2}(P - P^\top)$ is the discretised current — a $d \times d$ matrix assembled from the same batch cross-covariance a predictor/VICReg objective already forms. There is no separate Hodge solve in the gradient loop; the current is a by-product of training. (3) *The harmonic dimension is a spectral readout, not a simplicial one.* Each harmonic current appears as a conjugate pair of complex eigenvalues of P whose eigenfunctions wind; the count of *persistent* such pairs — stable under coarsening the pair-lag $\tau \rightarrow 2\tau$ and under encoder perturbation — estimates the active b_1 . That is an $O(d^3)$ eigendecomposition at evaluation time and an $O(d^2)$ assembly folded into the existing covariance computation, both negligible against the forward/backward pass; harmonic-versus-eddy is decided by this coarsening-persistence test rather than by constructing a complex. The honest limitation: this is a spectral *estimator* of the active harmonic dimension — exact only in the finite-graph case (E12), and in the continuum a cheap, graceful lower bound that can under-count modes too slow or too mixed to separate.

4. Certificates (E9, E10, E12, E13, E14, E15, E16)

Every claim is checked by a deterministic, exact, CPU-millisecond program, [empirical/apex_recovery/e9_irrev](#) wired into the suite’s pass/fail gate ([run_all.py](#)). There is no machine learning in it: the theory collapses both objectives to operator algebra on a circulant matrix, so the certificate is exact up to LAPACK ε . On the drift ring $\mathbb{Z}/12$ ($q = 0.5$, $b = 0.2$) it certifies, to machine precision:

claim	quantity	measured
blind-spot identity (2.2)	$\max_h \langle h, Th \rangle - \langle h, Sh \rangle $ over 202 encoders	3.3×10^{-16}
skew form vanishes (Lemma 1)	$\max_h \langle h, Ah \rangle $	2.1×10^{-17}
time-reversal degeneracy (Thm 1.1)	$\ S_{\text{fwd}} - S_{\text{rev}}\ _F$ / loss gap	1.9×10^{-16} / 8.9×10^{-16}
the worlds genuinely differ	$\ T_{\text{fwd}} - T_{\text{rev}}\ _F = 2\ A\ _F$	1.47
arrow in the spectrum	closed-form λ_1 vs numeric	4.5×10^{-16}
predictor block (Thm 2)	$\ B - [[\alpha, \beta], [-\beta, \alpha]]\ _F$	1.1×10^{-16}
predictor separates the arrow	$\ B_{\text{fwd}} - B_{\text{rev}}\ _F$	0.42 (> 0)
control: blind spot OFF at $q = b$	$\ A\ _F, \beta, \text{antisym}(B)$	0, 0, 0 (exact)

A sweep at fixed total rate $q + b$ confirms the picture in one image (the certificate’s figure): the single-encoder objective is **flat** across all drift asymmetries while the arrow $|\beta|$ rises linearly, and the predictor’s flow on the ring embedding spins forward or backward with the current. The same identity is verified on the conductor-blind-spot moduli $n \in \{6, 8, 12, 18, 30\}$ — it is literally the CBS ring with a current added.

The general non-normal form (E10) — [e10_nonnormal_svd.py](#), also in the gate — certifies §3.2 on a **conveyor**: a $\mathbb{Z}/n$ walk with position-dependent forward rates a_k (stationary law $\pi_k \propto 1/a_k$ in closed form), the maximally irreversible chain. It is genuinely non-normal ($\|TT^* - T^*T\| = 0.97$), Theorem 1’s identity still holds exactly (4×10^{-16}), and the recovered input and output charts **differ by up to 60°** (principal angles of the top singular functions), with the two-encoder optimum matching $\sum \sigma_i$ to 10^{-15} and beating every random whitened pair. The irreversibility gap is $\Delta = 0.40$ on the conveyor and $\Delta = 0.025$ on the normal drift ring (the phase deficit) — and **exactly 0** at detailed balance, where the chart split and the arrow vanish together. Time reversal swaps the left and right charts to 10^{-6} degrees. The contrast is the paper in one certificate: a current makes $\Delta > 0$; non-normality additionally splits the charts.

The topology (E12) — [e12_topological_blindspot.py](#), also in the gate — certifies Theorem 4 on four graphs of increasing cycle rank: a ring ($\beta_1 = 1$), a theta graph ($\beta_1 = 2$), the complete graph K_4 ($\beta_1 = 3$), and a 3×3 torus ($\beta_1 = 10$). For each it confirms β_1 three independent ways ($E - V + 1$, the nullity of the boundary operator, and the number of fundamental cycles, all agreeing), builds the β_1 circulations, and verifies that across the resulting β_1 -parameter family the single-encoder objective is **exactly invariant** (0 to machine precision) while the predictor recovers the current to 10^{-17} , that the antisymmetric part lives **entirely** in the cycle space (residual $\sim 10^{-17}$), and that the recovered current dimension **equals** β_1 in every case. The current is divergence-free (Kirchhoff residual $\sim 10^{-17}$), the gap Δ grows with it, and the detailed-balance control gives $\Delta = 0$ exactly. The dimension of the arrow is the cycle rank, measured. Finally E12 computes the **continuum refinement** of Theorem 5 by the Hodge 1-Laplacian $L_1 = d_1^\top d_1 + d_2 d_2^\top$: filling the torus’s square faces collapses $\beta_1 = 10$ to $b_1(T^2) = 2$, and filling the tetrahedral K_4 collapses $\beta_1 = 3$ to $b_1(S^2) = 0$.

(each a valid complex, $d_1 d_2 = 0$ exactly) — the mesh-dependent eddies fall away and only the topological modes remain.

Estimation in practice (E13) — [e13_estimating_the_arrow.py](#) — backs the remark above with numbers, and draws the practical line cleanly. The arrow’s *strength* Δ is estimable from finite sampled pairs and converges to the exact operator value (relative error $2.6\% \rightarrow 0.2\%$ as the sample grows from 10^3 to 10^5 pairs, variance shrinking as $1/\sqrt{m}$), at $O(d^3)$ eval-time cost — the in-loop-friendly signal, free from the predictor. Its *dimension* b_1 is **not** loop-cheap: a single-scale geometric (kNN flag-complex) estimate is wrong and scale-dependent (the torus’s b_1 is read as 47, 19, 13 for $k = 6, 9, 12$ against the true 2; the sphere’s as 31, 19, 9 against 0), and exact Vietoris–Rips persistence times out on a few hundred points offline. Topology is therefore an **offline, sub-sampled audit**, never an in-training computation — exactly as the overhead worry anticipates, and exactly why Δ (strength), not b_1 (count), is the quantity to track during learning.

The two axes are orthogonal (E14) — [e14_two_axes.py](#) — composes this paper’s reversibility axis with the parent’s cumulant axis on one product world (a warped Gaussian factor U and a drift ring V). Sweeping non-Gaussianity γ and irreversibility ρ , the linear-probe curvature ν_D depends on γ alone ($0 \rightarrow 0.27$) and the irreversibility gap Δ on ρ alone ($0 \rightarrow 0.02$), with **zero cross-talk to machine precision** — so all four corners of (Gaussian/non-Gaussian) $\times$ (reversible/irreversible) are realised independently. The arrow lives in the single product operator (its $L^2(\pi)$ -antisymmetric part grows with ρ and vanishes exactly at $\rho = 0$). A representation can be defeated on either axis or both; an audit that checks only linearity misses the arrow, and one that checks only reversibility misses the curved chart.

The offline audit, made efficient (E15) — [e15_efficient_betti.py](#) — turns Theorem 5’s b_1 from a prohibitive computation into a cheap one. The key is to read $b_1 = \dim \ker L_1$ as the count of **near-zero eigenvalues of the sparse Hodge 1-Laplacian**, obtained by shift-invert Lanczos that touches only the bottom of the spectrum — $O(\text{simplices} \times b_1)$ work, never the $O(E^3)$ dense boundary-rank E12 used. On complexes with known topology the nullity is *exact*: triangulated tori ($b_1 = 2$) scale to thousands of simplices in well under a second (where dense rank is hopeless), a triangulated sphere gives $b_1 = 0$ with no false positives, two disjoint tori give $b_1 = 4$. The honest residual is the *faithful sparse complex* from a raw point cloud: a single-scale flag complex has no good scale — a sampling gap breaks a loop (so a sampled circle reads $b_1 = 0$ at most k), larger k fills it, and a surface grows phantom loops (a Clifford-torus sample over-counts) — so the complex wants a persistence/alpha library on a sub-sample of the encoder embedding, after which this engine reads b_1 cheaply. The engine is solved here; E16 supplies the complex.

Closing the loop: the faithful complex (E16) — [e16_pointcloud_betti.py](#) — turns a raw point cloud into the right b_1 via **persistent homology** (GUDHI), the proper multi-scale answer to the scale problem E15 exposes. From samples of a circle, a flat Clifford torus, a sphere, and three circles it recovers $b_1 = 1, 2, 0, 3$ **exactly**, robustly (stable across sample size and additive noise $\sigma \approx 0.05$), at C++ speed (sub-second to \$2.6 s). The count uses a data-driven rule with no hand-tuned scale: b_1 is the number of H_1 bars clearing a dominance floor (a fixed fraction of the diameter), *confirmed* by a persistence-stability gap across that floor — one rule, four honest outcomes (all-noise $\rightarrow 0$, as the sphere correctly reads; all-signal $\rightarrow$ count; gap-confirmed $\rightarrow$ count; ambiguous $\rightarrow 0$). It composes exactly with E15: the faithful 2-skeleton at the persistence-chosen scale, handed to E15’s sparse-Hodge `hodge_b1`, returns the same $b_1 = 2$ on the Clifford torus. And it reconfirms the embedding lesson — the curved 3-D donut at matched N reads 0 because its uneven density raises the noise floor and collapses the gap, whereas the flat (encoder/diffusion) embedding is uniformly sampled and reads 2. (E16 needs a TDA library and so, like the trained certificates, is runnable but outside the dependency-light deterministic gate.)

5. Scope, and what comes next

The theorem is exact and assumption-light: it needs only a stationary pair and the $L^2(p)$ adjoint, and it holds for *every* encoder, not merely the optimum. Three honest boundaries. (i) **Normal, non-normal, topological, and continuum — settled (Theorems 3–5, E10/E12)**. On the symmetric ring the operator is normal and the arrow is the *imaginary spectrum*; a non-normal transition separates the **left and right singular charts** (§3.2, E10); on a graph the arrow is a β_1 -dimensional current space (§3.3, E12); and in the continuum it is the harmonic currents $H^1(M)$, of dimension $b_1(M)$ (§3.4, Theorem 5), to which the graph count refines as the mesh fills (torus $10 \rightarrow 2$). What remains genuinely open is **robust estimation from finite data**, and E13 maps it honestly: the arrow’s *strength* Δ is cheap and robust to estimate (it converges from sampled pairs at $O(d^3)$ eval cost, free from the predictor), but its *dimension* b_1 resists in-loop computation — single-scale geometric estimates are wrong and scale-dependent, and exact persistent homology times out on a few hundred points — so b_1 belongs to an offline, sub-sampled audit — and that audit is now end-to-end: **E16** supplies the *faithful complex* from raw samples by persistent homology (GUDHI), recovering b_1 exactly and robustly at C++ speed, and **E15** reads the *count* as the sparse-Hodge nullity, exact and scaling sub-second. The two compose (E16’s persistence-scale 2-skeleton $\rightarrow$ E15’s engine $\rightarrow$ the same b_1). What was the open practical frontier is closed for the cases studied; hardening it on large real encoders (sub-sampling strategy, the embedding’s effective dimension, confidence from persistence stability) is the natural engineering follow-up. (ii) **Population and trained — confirmed (E11)**. The theorems are about the objective and its optimum; E11 is the trained counterpart (the analogue of the parent’s E8), and it confirms them. Trained off-host by SGD, the single-encoder objective is **reversal-blind** — the loss gap and the Procrustes distance between a world and its time-reverse are both ≈ 0.01 (it converges to the same representation either way) — while the trained **predictor recovers the signed arrow**: $|\hat{\beta}| \approx 0.17$ against the ground-truth 0.15, flipping sign ($+0.17 \rightarrow -0.17$) under time reversal in a fixed gauge. SGD reaches the predictor optimum closely enough to read the current the single encoder cannot. (iii) **The two axes compose but are not the same**. Corollary 1 shows the reversibility blind spot sits *on top* of the cumulant blind spot: the recovered chart can be both arrow-erased (this paper) and curved-and-linearly-unreadable (parent). A representation can be defeated on either axis independently — measured directly in E14, where the curvature ν_D and the gap Δ vary with *zero cross-talk* across all four corners of (Gaussian/non-Gaussian) $\times$ (reversible/irreversible) — so an audit must check both.

The lesson for practice is sharp and architectural. If the world has an arrow — and video, language, and physical causality all do — a symmetric self-supervised objective will recover a time-reversal-blind representation no matter how much data or capacity it is given, because the arrow is not in the objective’s range. The fix is not scale; it is a predictor. The conductor blind spot taught that a second-order instrument cannot read an order-three fact; this is its sibling on the axis of time.

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